

James Thompson

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Education

Victoria University of Wellington Wellington, New Zealand
Master of Artificial Intelligence, A+ coursework, thesis not completed Feb 2025 – June 2026
Coursework focused on machine learning, deep learning.
Example projects include [LLM safety evaluations](#), [Deep RL for continuous robotic control algorithm design](#)
[Main project](#) is on AI alignment, specifically aligning LLMs to various subgroups of the New Zealand population.

Massey University Extramural, New Zealand
BSc in Computer Science, 94% A+ 2021 – 2023
Minoring in Statistics and Mathematics

Hutt Valley High School Lower Hutt, New Zealand
NCEA Level 3, Merit Endorsed 2020
Part of Premier Senoir debating team, and leader of the Snowsports team.

Work Experience

New Zealand Transport Accident Investigation Commission Wellington, NZ
Technical Lead (AI/Data Science) July 2023 – Current
Led the design and development of an AI-powered document analysis platform, deployed organisation-wide in November 2024 with a complete rewrite deployed in November 2025:

- Built an LLM-powered data pipeline that ingests and processes transport accident investigation reports from TAIC (NZ), ATSB (Australia), and TSB (Canada), extracting structured datasets of safety issues, recommendations, and report sections
- Implemented a RAG architecture with vector search (LanceDB) enabling semantic search across 100,000+ document segments, this powers an Agentic LLM to create a "smart assistant".
- Developed a web application featuring a smart assistant and search tool.

Open source: [TAIC-engine](#) | [TAIC-smart-tools](#)

Self-employed Remote, Global
Freelance Developer and Data Scientist April 2023 – Current
I work part-time on contract and freelance IT projects. My work includes developing business administration systems, such as a pupil attendance and invoicing system for a music school, and conducting data science projects like analyzing trends in health insurance claims.

Massey University Remote, NZ
Teaching Assistant February 2023 – November 2023
During my final year of undergraduate studies, I worked as a teaching assistant in Computer Science and Statistics courses. My responsibilities included grading assignments and tests, as well as providing one-on-one tutoring.

Telefund NZ Remote, NZ
Telefundraiser / Developer April 2021 – August 2023
I started as a tele-fundraiser and later transitioned into an administrative role. I developed an automated system to streamline administrative responsibilities and assist other callers and administrators.

Skills

Programming languages: Experienced with Python, comfortable with Javascript and bash, familiar with Rust and C++.

Machine Learning frameworks: Experienced with PyTorch and Jax/Flax

AI concepts: Experienced with LLMs (training, applications and evaluations), Deep RL and Evolutionary Computation

Devops: Experienced with Docker, Azure, Git and CI/CD pipelines

Research: Experienced working with HPC (Slurm), $\text{\LaTeX}$, academic writing and literature reviews

Soft: Experience in project planning, time management and self-directed work.

Extra-Curricular

Summer Research Massey University 2022/2023.....

Project: Worked with Dr. J Godfrey on the [BrailleR](#) package to enhance accessibility for vision-impaired users working with the R statistical programming language. Focused on:

- Expanded the graph explainer to include more ggplot2 graph types.
- Working with [Diagcess](#) (JS library) to create interactive webpages for blind users to explore ggplot graphs through a keyboard and sound interface

Competitive Snowboarding.....

Experience: I have been competing in [Freeride Snowboarding](#) since age 12, traveling overseas to compete with northern hemisphere athletes.

Highlights and achievements:

- Currently ranked 97th on the Global seeding list
- Won the 2025 Winter Games NZ North Face Frontier 4* event (held in August 2024)
- Competed at the Freeride Junior World Championship in 2020 (Austria) and 2021 (Switzerland).
- Qualified level 2 instructor with Snowboard Instruction New Zealand.

Scholarships and Awards

2025: Wellington Graduate Award, Victoria University

2023: Massey University Scholar Award

2023: Massey University Undergraduate Excellence Scholarship

2022: Massey School of Mathematical and Computational Sciences Summer Scholarship

2022: Massey University Elite Sports Bursary

2021: Massey University Academy of Sports Scholarship

2020: Upper Hutt Cosmopolitan Kickstart Career Scholarship

2020: Hutt Valley High School David Kaye Excellence Award

2019: Hutt Valley High School Best Senior Debater Award